

Introduction

During this lecture, the class discussed various **applications of Artificial Intelligence (AI)** with a focus on how AI is used in finance and real-world decision making. The discussion covered how AI systems analyze large amounts of data, help financial institutions detect fraud, and assist organizations in making better decisions.

AI in Finance

AI is increasingly used in the financial industry to improve efficiency and reduce risk. Banks and financial institutions use AI systems to evaluate credit risk and ensure compliance with financial regulations. Before approving a loan, banks analyze customer information, financial history, and credit scores using AI systems to determine whether the applicant qualifies for the loan.

AI systems can also analyze spending patterns and identify unusual behaviour. If a transaction occurs in an unusual location or appears suspicious, AI systems can flag the transaction for further investigation. This helps banks detect fraudulent activities quickly.

Fraud Detection and Anomaly Detection

Fraud prevention is one of the most important uses of AI in finance. AI systems monitor financial transactions and detect anomalies that may indicate fraud. For example, if a credit card is suddenly used in a different country or in an unusual location, AI systems can identify this as suspicious behavior.

Another example discussed in class was **credit card skimming**, where criminals copy credit card information at gas stations or other locations. AI systems can detect unusual transactions and alert financial institutions before more fraud occurs.

AI in Insurance and Claims Processing

AI is also used in insurance companies to verify claims. Insurance companies use AI systems to analyze documents and verify whether a claim is legitimate. For example, the system can check

medical records, death certificates, police reports, and other documents to ensure the information matches.

AI systems can also use data from online activity, such as cookies and browsing behaviour, to help assess risk levels for insurance policies.

Conclusion

Overall, AI plays an important role in modern financial systems. It helps organizations detect fraud, analyze financial risks, and process large amounts of data quickly. However, the lecture also emphasized that human oversight is still necessary when making financial decisions.